

BRUMLEY FINE, MO, USA

WMO: 724459

Lat: 38.096N

Lon: 92.553W

Elev: 869

StdP: 14.24

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 53931

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	5.2	10.4	-5.1	4.3	7.6	0.2	5.7	14.7	20.7	50.5	18.8	49.5	4.9	340	0.444

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	20.9	96.9	75.2	92.9	75.0	90.5	74.9	78.6	89.8	77.5	89.0	76.4	87.1	7.8	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
75.3	137.0	85.2	73.8	129.9	83.7	72.8	125.7	82.6	42.8	90.0	41.5	88.7	40.3	86.9	83.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
18.5	16.2	13.6	-1.5	100.2	6.5	3.9	-6.1	103.1	-9.9	105.4	-13.6	107.6	-18.3	110.4		
			-2.2	80.6	6.2	1.8	-6.7	81.9	-10.3	83.0	-13.8	84.0	-18.3	85.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	56.3	32.5	36.6	45.7	56.6	65.3	73.5	77.9	76.9	69.2	57.5	46.0	36.6
	DBStd	18.30	12.19	11.99	11.25	9.44	8.14	6.08	5.53	5.86	7.47	9.38	9.96	10.42
	HDD50	1872	556	399	215	41	3	0	0	0	0	35	192	431
	HDD65	4567	1009	795	603	281	102	11	1	2	41	270	572	880
	CDD50	4166	12	24	82	239	476	706	863	833	577	267	71	16
	CDD65	1384	0	0	5	29	109	266	399	370	168	36	2	0
	CDH74	13773	0	4	60	310	982	2582	4268	3690	1544	316	17	0
	CDH80	5832	0	0	6	61	286	1031	2040	1750	582	75	1	0
Wind	WSAvg	5.9	6.6	6.7	7.1	7.6	5.9	5.3	4.7	4.5	4.5	5.5	6.5	6.3
Precipitation	PrecAvg	45.6	2.3	2.4	3.4	5.0	6.0	4.8	4.5	3.9	4.7	3.2	3.1	2.4
	PrecMax	64.2	7.3	5.0	7.0	15.8	12.8	10.3	11.3	9.2	16.7	10.4	9.4	5.0
	PrecMin	30.8	0.6	0.2	1.4	0.8	1.3	1.5	1.0	1.8	1.0	0.4	0.4	0.8
	PrecStd	9.0	1.5	1.4	1.7	2.9	2.9	2.1	2.7	1.8	3.6	2.2	2.3	1.1
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	68.1	73.1	80.8	85.5	90.2	94.2	100.4	100.3	93.5	87.1	76.6	68.5
		MCWB	57.1	58.2	61.9	65.6	72.5	76.1	74.9	74.5	72.5	71.2	62.4	59.6
	2%	DB	62.9	66.3	74.8	81.4	86.3	91.0	96.9	97.0	90.3	81.6	71.7	63.0
		MCWB	52.8	54.7	57.8	63.1	71.3	75.3	75.7	74.9	73.6	66.5	58.0	54.6
	5%	DB	56.7	62.4	70.1	78.4	82.7	89.6	93.0	92.8	86.9	77.2	66.1	57.2
		MCWB	48.4	51.5	55.4	61.3	69.7	74.9	75.9	74.6	72.6	64.4	56.3	49.4
	10%	DB	51.6	56.5	64.2	73.1	80.6	86.4	90.5	90.0	82.2	72.8	62.9	53.6
		MCWB	44.3	47.2	52.3	59.3	67.6	73.6	75.8	74.3	69.5	62.4	53.7	46.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	59.7	61.0	64.1	68.4	75.6	79.3	80.4	80.7	77.4	73.3	64.0	60.6
		MCDB	66.6	69.9	77.5	80.7	85.2	89.5	91.4	90.7	87.9	82.6	71.7	67.4
	2%	WB	54.6	57.3	60.2	66.1	73.4	77.5	78.8	78.3	75.4	70.1	60.7	56.3
		MCDB	60.2	63.3	70.8	77.5	83.4	88.2	90.5	89.7	85.6	77.1	67.6	61.7
	5%	WB	49.3	52.8	57.2	63.8	71.3	76.0	77.8	77.0	73.8	67.1	58.3	51.4
		MCDB	55.9	60.2	67.1	73.5	80.6	85.8	89.5	88.5	83.7	74.2	64.8	55.9
	10%	WB	43.7	47.3	53.4	61.3	69.0	74.4	76.7	75.3	72.0	64.0	54.7	46.3
		MCDB	50.6	56.3	63.0	71.7	77.4	84.0	87.7	86.1	80.9	71.3	61.6	51.2
Mean Daily Temperature Range	5% DB	MDBR	18.9	19.9	21.9	22.2	21.2	20.2	20.9	21.4	21.8	21.7	20.1	17.7
		MCDBR	26.6	28.1	29.0	27.3	23.7	23.1	24.7	26.3	24.7	25.0	25.6	24.7
		MCWBR	18.4	18.7	16.5	14.5	12.3	10.6	9.0	9.6	10.9	13.4	16.6	17.3
	5% WB	MCDBR	22.8	24.8	24.4	22.3	21.3	20.5	21.1	21.4	20.9	20.7	21.0	21.1
		MCWBR	18.0	18.6	15.4	13.8	11.9	10.6	9.3	9.6	10.2	12.2	15.8	17.0
Clear-Sky Solar Irradiance	taub	0.301	0.313	0.341	0.388	0.421	0.431	0.453	0.457	0.390	0.346	0.309	0.296	
	taud	2.477	2.443	2.418	2.291	2.262	2.265	2.226	2.214	2.373	2.486	2.535	2.517	
	Ebn at Noon	280	291	292	282	274	270	263	259	272	276	276	275	
	Edh at Noon	25	30	34	41	43	43	44	44	35	29	24	23	
All-Sky Solar Radiation	RadAvg	705	913	1219	1584	1797	2043	2001	1780	1536	1105	765	601	
	RadStd	78	106	104	99	159	165	135	105	106	119	80	64	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (17 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page